

SMART ADC SELECTION

Can Simplify And Improve Design

There are so many standalone and integrated analogue-to-digital converters available today. Which ADC is best for your application? How important is sampling rate or architecture type? How can you pick a cost- and time-efficient ADC?



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Converters rule the world of embedded systems as they communicate with real-world signals, starting from a sensor that converts data (temperature, pressure, etc) to analogue voltage/current output. An analogue-to-digital converter (ADC) transforms the analogue output to digital form for digital systems to read and process.

Considering the varieties and price range, finding the best ADC for your application may be a difficult task. While

the first step is to look for speed, resolution, and accuracy, here are a few more parameters to check before selecting the ADC.

What to look for

Analogue-to-digital conversion involves two steps—sampling and quantisation. Inside an ADC, a switch capacitor takes snapshots of the input signal at regular intervals (known as sampling frequency). These snapshot values are rounded off to the nearest value for digital representation.

Resolution is the first factor to consider here. It refers to the number of output bits that the ADC can generate per conversion and determines the smallest input signal that the ADC can represent.

Then comes accuracy—the close matching of the input to the desired output. Third is the sampling rate or speed, which is decided based on how fast the input signal is changing or input signal bandwidth.

A perfect reconstruction is possible only if sampling rate is equal to or greater than twice the highest frequency in the original signal. This minimum required sampling rate is known as the Nyquist rate. Though this little detail sounds harmless, it can create a major challenge between theoretical sampling and analogue-to-digital conversion. This can be avoided by band limiting the input signal to meet Nyquist rate.

In quantisation, it is important to have a very low quantisation error; even an ideal ADC will have some minimal noise. This can be achieved by increasing the number of levels or the resolution of ADC,

Resolution, Bits (n)	2^n	LSB, mV (2.5V FS)	% Full Scale	ppm Full Scale	dB Full Scale
8	256	9.77	0.391	3906	-48.0
10	1024	2.44	0.098	977	-60.0
12	4096	0.610	0.024	244	-72.0
14	16,384	0.153	0.006	61	-84.0
16	65,536	0.038	0.0015	15	-96.0
18	262,144	0.0095	0.00038	3.8	-108.0

Fig. 1: ADC resolution vs quantisation parameters

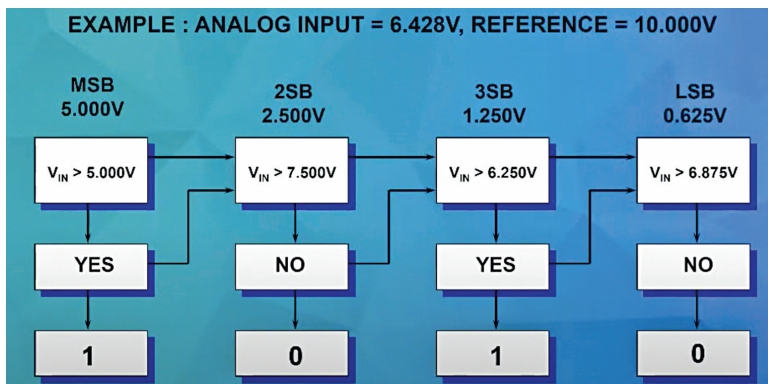


Fig. 2: Working of SAR ADC